Contents

1 Orbitals And Their Symmetry

We use the following ordering of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{rl} b_{1u} \;\text{—} & \text{—}\; a_u \\ b_{2g} \;\text{—} & \text{—}\; b_{3g} \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:

$$\begin{array}{cc} \text{—} & \text{—} \\ \text{—} & \text{—} \end{array}$$

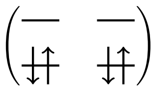
where the character of the orbital is defined by its position.

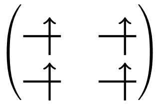
Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital ordering within the group of monomer orbitals follows the above mentioned definition.

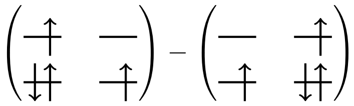
The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

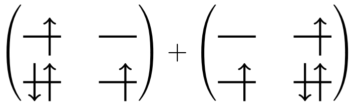
$$\begin{aligned} a_u^l &= +b_{1g} + a_u \\ b_{1u}^l &= +a_g + b_{1u} \\ b_{2g}^l &= +b_{3u} + b_{2g} \\ b_{3g}^l &= +b_{2u} + b_{3g} \\ a_u^r &= -b_{1g} + a_u \\ b_{1u}^r &= -a_g + b_{1u} \\ b_{2g}^r &= -b_{3u} + b_{2g} \\ b_{3g}^r &= -b_{2u} + b_{3g} \end{aligned}$$

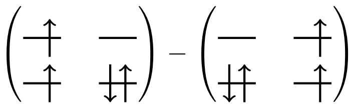
2 Monomer States and Configurations

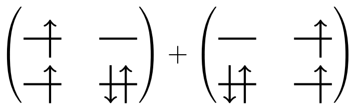
$S = \quad +$

$$= \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^0}_{a_g}\right|$$

$Q = \quad +$

$$= \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^1}_{a_g}\right|$$

$i^3b_{2u} = \quad +$

$$= \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b_{2u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b_{2u}}\right|$$

$e^3b_{2u} = \quad +$

$$= \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b_{2u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b_{2u}}\right|$$

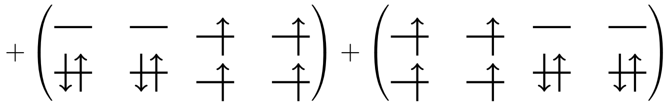
$e^3b_{3u} = \quad +$

$$= \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b_{3u}}\right|$$

$i^3b_{3u} = \quad +$

$$= \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b_{3u}}\right|$$

3 Linear Combinations

3.1 $S \otimes Q + Q \otimes S$

$$S \otimes Q + Q \otimes S$$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^0}_{a^l_g}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^1}_{a^r_g}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^1}_{a^l_g}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^0}_{a^r_g}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^2(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^0(+b_{1g}+a_u)^0}_{a^l_g}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^1(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^1}_{a^r_g}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

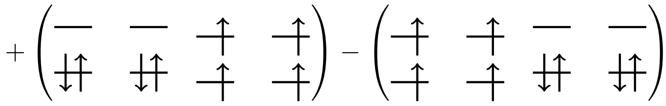
$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

3.2 $\boldsymbol{S \otimes Q - Q \otimes S}$

$$S \otimes Q - Q \otimes S$$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^0}_{a^l_g}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^1}_{a^r_g}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^1}_{a^l_g}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^0}_{a^r_g}\right|$$

substitution by dimer orbitals:

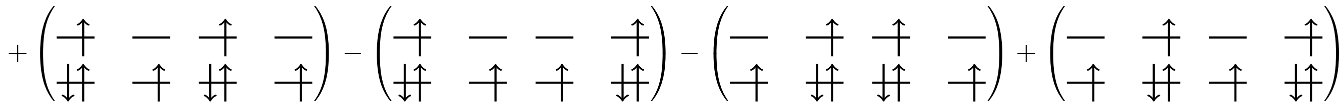
$$+\left|\underbrace{(+b_{2u}+b_{3g})^2(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^0(+b_{1g}+a_u)^0}_{a^l_g}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^1(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^1}_{a^r_g}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$\begin{aligned}
&-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}}\right| \\
&-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}}\right| \\
&-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}}\right| \\
&-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}}\right|
\end{aligned}$$

3.3 $i^3b_{2u} \otimes i^3b_{2u}$

$$i^3b_{2u} \otimes i^3b_{2u}$$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^1(-b_{3u}+b_{2g})^2(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{2u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + \left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}\right|$$

3.4 $i^3 b_{2u} \otimes e^3 b_{2u} + e^3 b_{2u} \otimes i^3 b_{2u}$

$$i^3 b_{2u} \otimes e^3 b_{2u} + e^3 b_{2u} \otimes i^3 b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$
$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$
$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$
$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$
$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

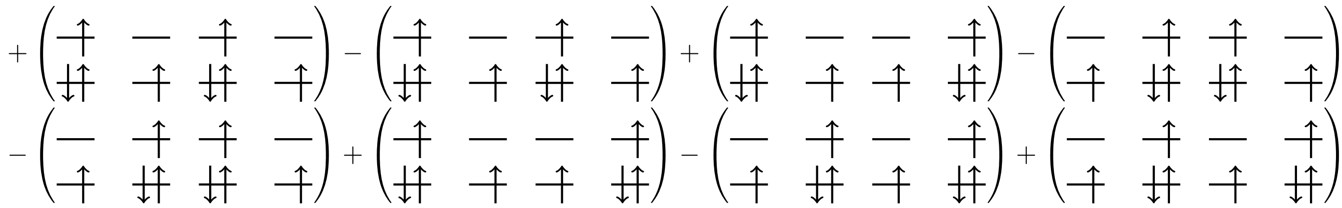
$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^1(-b_{3u}+b_{2g})^2(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{2u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}\right|$$

3.5 $i^3 b_{2u} \otimes e^3 b_{2u} - e^3 b_{2u} \otimes i^3 b_{2u}$

$i^3 b_{2u} \otimes e^3 b_{2u} - e^3 b_{2u} \otimes i^3 b_{2u}$



monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| \\
&+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| \\
&-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right| \\
&-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|
\end{aligned}
$$

substitution by dimer orbitals:

$$
+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^1(-b_{3u}+b_{2g})^2(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{2u}}\right| + \dots
$$

(multiplied out and summed up) dimer ci vectors:

$$
\begin{aligned}
&+4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}}\right| - 4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}}\right| \\
&+4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}}\right| - 4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}}\right| \\
&-4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}}\right| + 4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}}\right| \\
&-4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}}\right| + 4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}}\right|
\end{aligned}
$$

3.6 $i^3 b_{2u} \otimes e^3 b_{3u} + e^3 b_{3u} \otimes i^3 b_{2u}$

$$i^3 b_{2u} \otimes e^3 b_{3u} + e^3 b_{3u} \otimes i^3 b_{2u}$$

$$
\begin{aligned}
&+\begin{pmatrix} \uparrow & \text{—} & \uparrow & \text{—} \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
+\begin{pmatrix} \uparrow & \text{—} & \uparrow & \text{—} \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
-\begin{pmatrix} \uparrow & \text{—} & \text{—} & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
-\begin{pmatrix} \text{—} & \uparrow & \uparrow & \text{—} \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\
&-\begin{pmatrix} \text{—} & \uparrow & \uparrow & \text{—} \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
-\begin{pmatrix} \uparrow & \text{—} & \text{—} & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
+\begin{pmatrix} \text{—} & \uparrow & \text{—} & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}
+\begin{pmatrix} \text{—} & \uparrow & \text{—} & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}
\end{aligned}
$$

monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| \\
&-\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| \\
&-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right| \\
&+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|
\end{aligned}
$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$
\begin{aligned}
&+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right| \\
&-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right| \\
&+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right| \\
&-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right|
\end{aligned}
$$

3.7 $i^3 b_{2u} \otimes e^3 b_{3u} - e^3 b_{3u} \otimes i^3 b_{2u}$

$$i^3 b_{2u} \otimes e^3 b_{3u} - e^3 b_{3u} \otimes i^3 b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}\right|$$

3.8 $i^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes i^3b_{2u}$

$$i^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes i^3b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$
$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$
$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$
$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$
$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

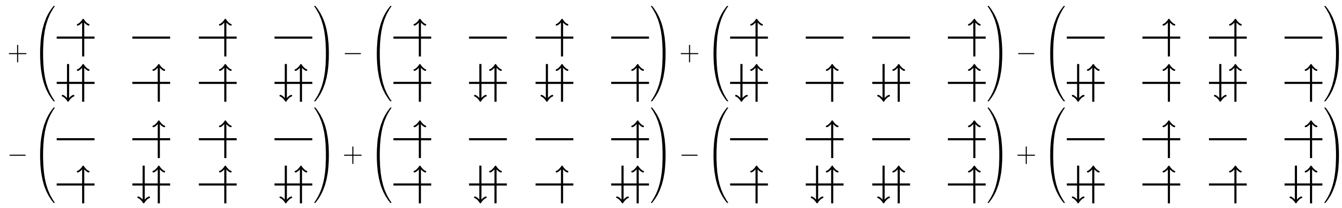
$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right|$$
$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right|$$
$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right|$$
$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right|$$

3.9 $i^3 b_{2u} \otimes i^3 b_{3u} - i^3 b_{3u} \otimes i^3 b_{2u}$

$$i^3 b_{2u} \otimes i^3 b_{3u} - i^3 b_{3u} \otimes i^3 b_{2u}$$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u}\right|$$

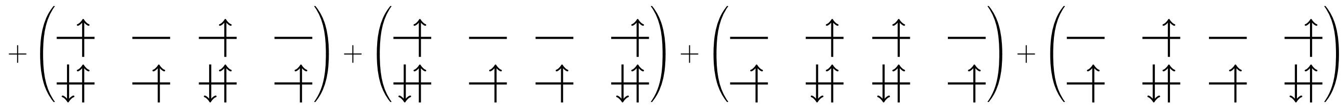
$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}\right|$$

3.10 $e^3 b_{2u} \otimes e^3 b_{2u}$

$e^3 b_{2u} \otimes e^3 b_{2u}$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^1(-b_{3u}+b_{2g})^2(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{2u}}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$+\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

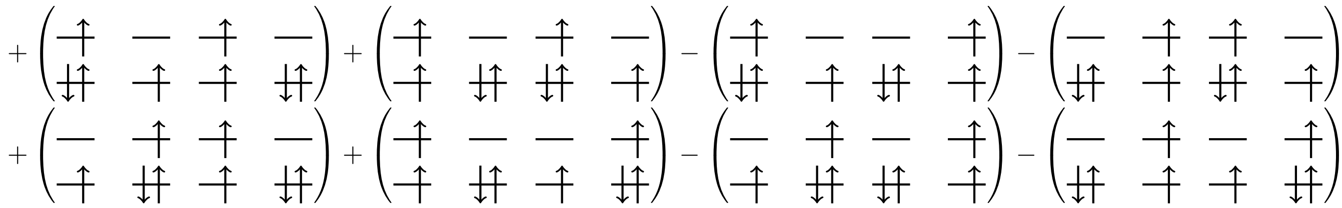
$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + \left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}\right|$$

3.11 $e^3b_{2u} \otimes e^3b_{3u} + e^3b_{3u} \otimes e^3b_{2u}$

$$e^3b_{2u} \otimes e^3b_{3u} + e^3b_{3u} \otimes e^3b_{2u}$$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right|$$

3.12 $e^3b_{2u} \otimes e^3b_{3u} - e^3b_{3u} \otimes e^3b_{2u}$

$$e^3b_{2u} \otimes e^3b_{3u} - e^3b_{3u} \otimes e^3b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$+\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u}\right|$$

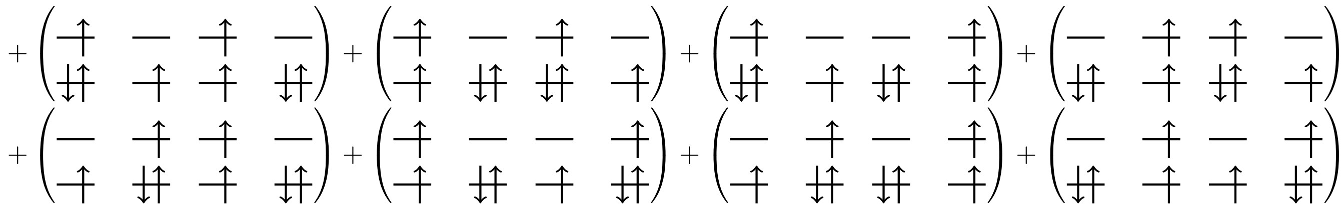
$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}\right|$$

3.13 $e^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{2u}$

$$e^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{2u}$$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{b_{1g}}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1g}}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1g}}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{b_{1g}}\right|$$

3.14 $e^3b_{2u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{2u}$

$$e^3b_{2u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{2u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

$$+\begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^1(a^l_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^1(a^r_u)^0}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^0(a^l_u)^1}_{b^l_{2u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^0(a^r_u)^1}_{b^r_{2u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^1(+b_{3u}+b_{2g})^2(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{2u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u} + 2\cdot \underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_u}\right|$$

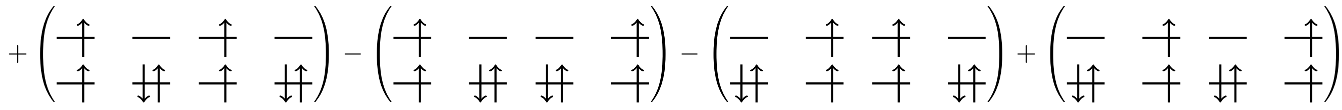
$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u} + 2\cdot \underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_u} + 2\cdot \underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_u}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u} + 2\cdot \underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_u}\right|$$

3.15 $e^3 b_{3u} \otimes e^3 b_{3u}$

$e^3 b_{3u} \otimes e^3 b_{3u}$



monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right|$$

$$-\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right|$$

substitution by dimer orbitals:

$$+\left|\underbrace{(+b_{2u}+b_{3g})^2(+b_{3u}+b_{2g})^1(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$+\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right|$$

$$+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right|$$

$$-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + \left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}\right|$$

3.16 $e^3b_{3u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{3u}$

$$e^3b_{3u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{3u}$$

$$+\begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$
$$-\begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix}$$

monomer ci vectors:

$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right|$$
$$+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right|$$
$$-\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right|$$
$$-\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| - \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right|$$

substitution by dimer orbitals:

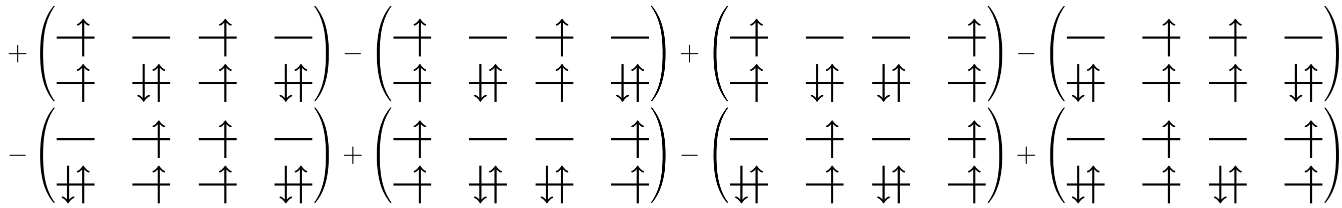
$$+\left|\underbrace{(+b_{2u}+b_{3g})^2(+b_{3u}+b_{2g})^1(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \ldots$$

(multiplied out and summed up) dimer ci vectors:

$$+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}\right|$$

3.17 $e^3b_{3u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{3u}$

$$e^3b_{3u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{3u}$$



monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r}\right| - \left|\underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r}\right| \\
&+\left|\underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r}\right| - \left|\underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r}\right| \\
&-\left|\underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^2(b_{2g}^r)^1(b_{1u}^r)^1(a_u^r)^0}_{b_{3u}^r}\right| + \left|\underbrace{(b_{3g}^l)^2(b_{2g}^l)^1(b_{1u}^l)^1(a_u^l)^0}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r}\right| \\
&-\left|\underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r}\right| + \left|\underbrace{(b_{3g}^l)^1(b_{2g}^l)^2(b_{1u}^l)^0(a_u^l)^1}_{b_{3u}^l}\right| \otimes \left|\underbrace{(b_{3g}^r)^1(b_{2g}^r)^2(b_{1u}^r)^0(a_u^r)^1}_{b_{3u}^r}\right|
\end{aligned}
$$

substitution by dimer orbitals:

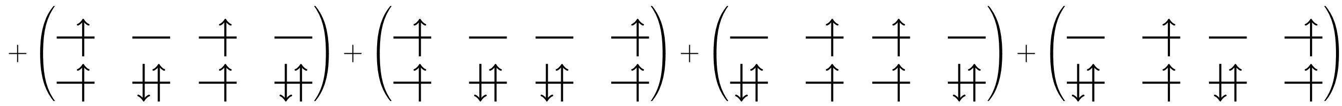
$$+\left|\underbrace{(+b_{2u}+b_{3g})^2(+b_{3u}+b_{2g})^1(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b_{3u}^l}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b_{3u}^r}\right| + \dots$$

(multiplied out and summed up) dimer ci vectors:

$$
\begin{aligned}
&-4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}}\right| + 4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}}\right| \\
&+4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}}\right| - 4\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}}\right| \\
&-4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{b_{1u}}\right| + 4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{b_{1u}}\right| \\
&+4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{b_{1u}}\right| - 4\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{b_{1u}}\right|
\end{aligned}
$$

3.18 $i^3 b_{3u} \otimes i^3 b_{3u}$

$i^3 b_{3u} \otimes i^3 b_{3u}$



monomer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^2(b^l_{2g})^1(b^l_{1u})^1(a^l_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right| \\
&+\left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^2(b^r_{2g})^1(b^r_{1u})^1(a^r_u)^0}_{b^r_{3u}}\right| + \left|\underbrace{(b^l_{3g})^1(b^l_{2g})^2(b^l_{1u})^0(a^l_u)^1}_{b^l_{3u}}\right| \otimes \left|\underbrace{(b^r_{3g})^1(b^r_{2g})^2(b^r_{1u})^0(a^r_u)^1}_{b^r_{3u}}\right|
\end{aligned}
$$

substitution by dimer orbitals:

$$
+\left|\underbrace{(+b_{2u}+b_{3g})^2(+b_{3u}+b_{2g})^1(+a_g+b_{1u})^1(+b_{1g}+a_u)^0}_{b^l_{3u}}\right| \otimes \left|\underbrace{(-b_{2u}+b_{3g})^2(-b_{3u}+b_{2g})^1(-a_g+b_{1u})^1(-b_{1g}+a_u)^0}_{b^r_{3u}}\right| + \dots
$$

(multiplied out and summed up) dimer ci vectors:

$$
\begin{aligned}
&+\left|\underbrace{(b_{2u})^2(b_{3g})^2(b_{3u})^1(b_{2g})^1(a_g)^1(b_{1u})^1(b_{1g})^0(a_u)^0}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right| \\
&-2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right| \\
&+2\cdot\left|\underbrace{(b_{2u})^2(b_{3g})^1(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| + 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^1(b_{1u})^0(b_{1g})^1(a_u)^0}_{a_g}\right| \\
&-2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^1(b_{2g})^2(a_g)^0(b_{1u})^1(b_{1g})^0(a_u)^1}_{a_g}\right| - 2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^1(b_{1u})^0(b_{1g})^0(a_u)^1}_{a_g}\right| \\
&+2\cdot\left|\underbrace{(b_{2u})^1(b_{3g})^2(b_{3u})^2(b_{2g})^1(a_g)^0(b_{1u})^1(b_{1g})^1(a_u)^0}_{a_g}\right| + \left|\underbrace{(b_{2u})^1(b_{3g})^1(b_{3u})^2(b_{2g})^2(a_g)^0(b_{1u})^0(b_{1g})^1(a_u)^1}_{a_g}\right|
\end{aligned}
$$

4 Symmetry Conclusion

Dimer State	Symmetry
$S \otimes Q + Q \otimes S$	a_g
$S \otimes Q - Q \otimes S$	b_{1u}
$i^3b_{2u} \otimes i^3b_{2u}$	a_g
$i^3b_{2u} \otimes e^3b_{2u} + e^3b_{2u} \otimes i^3b_{2u}$	a_g
$i^3b_{2u} \otimes e^3b_{2u} - e^3b_{2u} \otimes i^3b_{2u}$	b_{1u}
$i^3b_{2u} \otimes e^3b_{3u} + e^3b_{3u} \otimes i^3b_{2u}$	b_{1g}
$i^3b_{2u} \otimes e^3b_{3u} - e^3b_{3u} \otimes i^3b_{2u}$	a_u
$i^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes i^3b_{2u}$	b_{1g}
$i^3b_{2u} \otimes i^3b_{3u} - i^3b_{3u} \otimes i^3b_{2u}$	a_u
$e^3b_{2u} \otimes e^3b_{2u}$	a_g
$e^3b_{2u} \otimes e^3b_{3u} + e^3b_{3u} \otimes e^3b_{2u}$	b_{1g}
$e^3b_{2u} \otimes e^3b_{3u} - e^3b_{3u} \otimes e^3b_{2u}$	a_u
$e^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{2u}$	b_{1g}
$e^3b_{2u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{2u}$	a_u
$e^3b_{3u} \otimes e^3b_{3u}$	a_g
$e^3b_{3u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{3u}$	a_g
$e^3b_{3u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{3u}$	b_{1u}
$i^3b_{3u} \otimes i^3b_{3u}$	a_g

$\Rightarrow$ 7x a_g and 3x b_{1u} and 4x b_{1g} and 4x a_u

5 CI Vector Conclusion

Dimer States and Their CI Vectors:

- $\boldsymbol{S \otimes Q + Q \otimes S}$:

$$+2 \cdot |2a2a0a0a| + 2 \cdot |2a2aa0a0| + 2 \cdot |2aa20aa0| + 2 \cdot |2aa2a00a| + 2 \cdot |a22a0aa0| + 2 \cdot |a22aa00a|$$

$$+2 \cdot |a2a20a0a| + 2 \cdot |a2a2a0a0|$$

- $\boldsymbol{S \otimes Q - Q \otimes S}$:

$$-2 \cdot |2a2a0aa0| - 2 \cdot |2a2aa00a| - 2 \cdot |2aa20a0a| - 2 \cdot |2aa2a0a0| - 2 \cdot |a22a0a0a| - 2 \cdot |a22aa0a0|$$

$$-2 \cdot |a2a20aa0| - 2 \cdot |a2a2a00a|$$

- $\boldsymbol{i^3b_{2u} \otimes i^3b_{2u}}$:

$$+ |aa22aa00| + 2 \cdot |a22aa00a| - 2 \cdot |a22a0aa0| + 2 \cdot |a2a2a0a0| - 2 \cdot |a2a20a0a| - 2 \cdot |2a2aa0a0|$$

$$+2 \cdot |2a2a0a0a| - 2 \cdot |2aa2a00a| + 2 \cdot |2aa20aa0| + |22aa00aa|$$

- $\boldsymbol{i^3b_{2u} \otimes e^3b_{2u} + e^3b_{2u} \otimes i^3b_{2u}}$:

$$+2 \cdot |aa22aa00| - 2 \cdot |22aa00aa|$$

- $\boldsymbol{i^3b_{2u} \otimes e^3b_{2u} - e^3b_{2u} \otimes i^3b_{2u}}$:

$$+4 \cdot |a22aa0a0| - 4 \cdot |a22a0a0a| + 4 \cdot |a2a2a00a| - 4 \cdot |a2a20aa0| - 4 \cdot |2a2aa00a| + 4 \cdot |2a2a0aa0|$$

$$-4 \cdot |2aa2a0a0| + 4 \cdot |2aa20a0a|$$

- $\boldsymbol{i^3b_{2u} \otimes e^3b_{3u} + e^3b_{3u} \otimes i^3b_{2u}}$:

$$+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| - 2 \cdot |aa22a00a| + 2 \cdot |aa220aa0| + 2 \cdot |22aa0aa0| - 2 \cdot |22aaa00a|$$

$$-2 \cdot |2aa200aa| + 2 \cdot |a22a00aa|$$

- $\boldsymbol{i^3b_{2u} \otimes e^3b_{3u} - e^3b_{3u} \otimes i^3b_{2u}}$:

$$-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| + 2 \cdot |aa22a0a0| - 2 \cdot |aa220a0a| + 2 \cdot |22aa0a0a| - 2 \cdot |22aaa0a0|$$

$$-2 \cdot |2a2a00aa| + 2 \cdot |a2a200aa|$$

- $\boldsymbol{i^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes i^3b_{2u}}$:

$$+2 \cdot |a22aaa00| - 2 \cdot |2aa2aa00| + 2 \cdot |aa22a00a| - 2 \cdot |aa220aa0| + 2 \cdot |22aa0aa0| - 2 \cdot |22aaa00a|$$

$$+2 \cdot |2aa200aa| - 2 \cdot |a22a00aa|$$

- $\boldsymbol{i^3b_{2u} \otimes i^3b_{3u} - i^3b_{3u} \otimes i^3b_{2u}}$:

$$-2 \cdot |a2a2aa00| + 2 \cdot |2a2aaa00| - 2 \cdot |aa22a0a0| + 2 \cdot |aa220a0a| + 2 \cdot |22aa0a0a| - 2 \cdot |22aaa0a0|$$

$$+2 \cdot |2a2a00aa| - 2 \cdot |a2a200aa|$$

- $\boldsymbol{e^3b_{2u} \otimes e^3b_{2u}}$:

$$+ |aa22aa00| - 2 \cdot |a22aa00a| + 2 \cdot |a22a0aa0| - 2 \cdot |a2a2a0a0| + 2 \cdot |a2a20a0a| + 2 \cdot |2a2aa0a0|$$

$$-2 \cdot |2a2a0a0a| + 2 \cdot |2aa2a00a| - 2 \cdot |2aa20aa0| + |22aa00aa|$$

- $\boldsymbol{e^3b_{2u} \otimes e^3b_{3u} + e^3b_{3u} \otimes e^3b_{2u}}$:

$$+2\cdot|a22aaa00| - 2\cdot|2aa2aa00| - 2\cdot|aa22a00a| + 2\cdot|aa220aa0| - 2\cdot|22aa0aa0| + 2\cdot|22aaa00a|$$

$$+2\cdot|2aa200aa| - 2\cdot|a22a00aa|$$

- $\boldsymbol{e^3b_{2u} \otimes e^3b_{3u} - e^3b_{3u} \otimes e^3b_{2u}}$:

$$-2\cdot|a2a2aa00| + 2\cdot|2a2aaa00| + 2\cdot|aa22a0a0| - 2\cdot|aa220a0a| - 2\cdot|22aa0a0a| + 2\cdot|22aaa0a0|$$

$$+2\cdot|2a2a00aa| - 2\cdot|a2a200aa|$$

- $\boldsymbol{e^3b_{2u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{2u}}$:

$$+2\cdot|a22aaa00| - 2\cdot|2aa2aa00| + 2\cdot|aa22a00a| - 2\cdot|aa220aa0| - 2\cdot|22aa0aa0| + 2\cdot|22aaa00a|$$

$$-2\cdot|2aa200aa| + 2\cdot|a22a00aa|$$

- $\boldsymbol{e^3b_{2u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{2u}}$:

$$-2\cdot|a2a2aa00| + 2\cdot|2a2aaa00| - 2\cdot|aa22a0a0| + 2\cdot|aa220a0a| - 2\cdot|22aa0a0a| + 2\cdot|22aaa0a0|$$

$$-2\cdot|2a2a00aa| + 2\cdot|a2a200aa|$$

- $\boldsymbol{e^3b_{3u} \otimes e^3b_{3u}}$:

$$+|22aaaa00| - 2\cdot|2aa2a00a| + 2\cdot|2aa20aa0| + 2\cdot|2a2aa0a0| - 2\cdot|2a2a0a0a| - 2\cdot|a2a2a0a0|$$

$$+2\cdot|a2a20a0a| + 2\cdot|a22aa00a| - 2\cdot|a22a0aa0| + |aa2200aa|$$

- $\boldsymbol{e^3b_{3u} \otimes i^3b_{3u} + i^3b_{3u} \otimes e^3b_{3u}}$:

$$+2\cdot|22aaaa00| - 2\cdot|aa2200aa|$$

- $\boldsymbol{e^3b_{3u} \otimes i^3b_{3u} - i^3b_{3u} \otimes e^3b_{3u}}$:

$$-4\cdot|2aa2a0a0| + 4\cdot|2aa20a0a| + 4\cdot|2a2aa00a| - 4\cdot|2a2a0aa0| - 4\cdot|a2a2a00a| + 4\cdot|a2a20aa0|$$

$$+4\cdot|a22aa0a0| - 4\cdot|a22a0a0a|$$

- $\boldsymbol{i^3b_{3u} \otimes i^3b_{3u}}$:

$$+|22aaaa00| + 2\cdot|2aa2a00a| - 2\cdot|2aa20aa0| - 2\cdot|2a2aa0a0| + 2\cdot|2a2a0a0a| + 2\cdot|a2a2a0a0|$$

$$-2\cdot|a2a20a0a| - 2\cdot|a22aa00a| + 2\cdot|a22a0aa0| + |aa2200aa|$$